



MATHEMATICS (EXTENSION 1)

2013 HSC Course Assessment Task 3 (Trial Examination)

June 26, 2013

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 9)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Mr Lam

☐ 12M4A – Mr Fletcher

☐ 12M3B – Mr Berry

☐ 12M4B – Ms Beevers

☐ 12M3C – Mr Lin

☐ 12M4C – Ms Ziazaris

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\frac{\quad}{10}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{70}$	

Section I

10 marks

Attempt Question 1 to 10

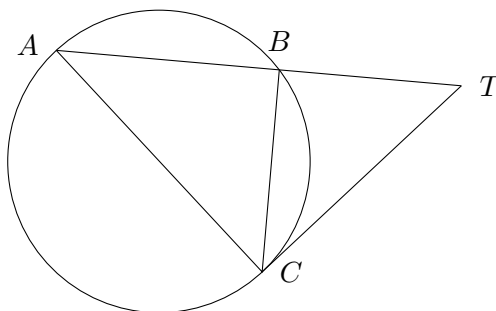
Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions

Marks

1. What is the acute angle between the lines $y = 2x - 3$ and $3x + 5y - 1 = 0$, correct to the nearest degree? 1
 (A) 32° (B) 50° (C) 82° (D) 86°
2. The coordinates of the points that divide the interval joining $(-7, 5)$ and $(-1, -7)$ externally in the ratio $1 : 3$ are 1
 (A) $(-10, 8)$ (B) $(-10, 11)$ (C) $(2, 8)$ (D) $(2, 11)$
3. CT is a tangent to the circle. AB is a secant, intersecting the circle at A and B . AB intersects CT at T . 1



Which of the following statements is correct?

- (A) $CT^2 = AC \times BC$ (C) $CT^2 = AC \times BT$
 - (B) $CT^2 = AB \times BC$ (D) $CT^2 = AT \times BT$
4. What is the domain and range of $y = \cos^{-1}\left(\frac{3x}{2}\right)$? 1
- (A) $D = \left\{x : -\frac{2}{3} \leq x \leq \frac{2}{3}\right\}, R = \{y : 0 \leq y \leq \pi\}$
 - (B) $D = \{x : -1 \leq x \leq 1\}, R = \{y : 0 \leq y \leq \pi\}$
 - (C) $D = \left\{x : -\frac{2}{3} \leq x \leq \frac{2}{3}\right\}, R = \{y : -\pi \leq y \leq \pi\}$
 - (D) $D = \{x : -1 \leq x \leq 1\}, R = \{y : -\pi \leq y \leq \pi\}$

5. Which of the following is equivalent to $\int \frac{dx}{4x^2 + 9}$, ignoring the constant of integration? 1
- (A) $\tan^{-1} \frac{2x}{3}$ (C) $\frac{2}{3} \tan^{-1} \frac{2x}{3}$
- (B) $\frac{1}{6} \tan^{-1} \frac{2x}{3}$ (D) $\frac{3}{2} \tan^{-1} \frac{2x}{3}$
6. Which of the following is the general solution to $3 \tan^2 x - 1 = 0$? 1
- (A) $n\pi \pm \frac{\pi}{6}$ (C) $n\pi \pm \frac{\pi}{3}$
- (B) $2n\pi \pm \frac{\pi}{6}$ (D) $2n\pi \pm \frac{\pi}{3}$
7. What is the value of $\sin(\alpha + \beta)$ if $\sin \alpha = \frac{8}{17}$ and $\sin \beta = \frac{4}{5}$? 1
- (A) $\frac{36}{85}$ (C) $\frac{84}{85}$
- (B) $\frac{77}{85}$ (D) $\frac{88}{85}$
8. Which of the following represents the exact value of $\int_0^{\frac{\pi}{8}} \cos^2 x \, dx$? 1
- (A) $\frac{\pi - 2\sqrt{2}}{16}$ (C) $\frac{\pi + 2\sqrt{2}}{16}$
- (B) $\frac{\pi - 2\sqrt{2}}{8}$ (D) $\frac{\pi + 2\sqrt{2}}{8}$
9. Which of the following represents the derivative of $y = \cos^{-1} \left(\frac{1}{x} \right)$? 1
- (A) $-\frac{1}{\sqrt{x^2 - 1}}$ (C) $\frac{1}{\sqrt{x^2 - 1}}$
- (B) $-\frac{1}{x\sqrt{x^2 - 1}}$ (D) $\frac{1}{x\sqrt{x^2 - 1}}$
10. Which of the following expressions is true? 1
- (A) $\tan^{-1} x = \sin^{-1} \frac{1}{\sqrt{1 - x^2}}$ (C) $\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1 - x^2}}$
- (B) $\tan^{-1} x = \sin^{-1} \frac{1}{\sqrt{1 + x^2}}$ (D) $\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1 + x^2}}$

Examination continues overleaf...

Section II

60 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Solve for x : $\frac{5}{x-1} > 2$		2
(b) Find the value of θ , such that		4
$\sqrt{3} \cos \theta - \sin \theta = 1$		
where $0 \leq \theta \leq 2\pi$.		
(c) A freshly baked cake is cooling in a room of constant temperature of 20°C . At time t minutes, its temperature T decreases according to the equation		
$\frac{dT}{dt} = -k(T - 20)$		
where k is a positive constant.		
The initial temperature of the cake is 150°C and it cools to 100°C after 15 minutes.		
i. Show that $T = 20 + Ae^{-kt}$ (where A is a constant) is a solution to this equation.		1
ii. Find the values A and k , giving k correct to 3 decimal places.		2
iii. How long will it take for the cake to cool to 25°C ? (Use the value of k obtained in the previous part)		2
(d) A particle is moving in a straight line. After time t seconds, it has displacement x metres from a fixed point O on the line. Its velocity $v \text{ ms}^{-1}$ is given by		
$v = \sqrt{x}$		
Initially, the particle is 1 m to the right of O .		
i. Show that its acceleration a is independent of x .		1
ii. Express x in terms of t .		2
iii. Find the distance travelled by the particle during the third second of its motion.		1

Question 12 (15 Marks)

Commence a NEW page.

Marks

(a) Show that $\frac{\sin x}{1 - \cos x} = \cot \frac{x}{2}$. **2**

(b) Use the substitution $u = \sin^2 x$ to evaluate **4**

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin 2x}{1 + \sin^2 x} dx$$

Give your answer in simplest exact form.

(c) Let $f(x) = x^2 - 2x$ for $x \geq 1$. **3**

i. On the same set of axes, sketch the graphs of $y = f(x)$, $y = x$ and the inverse $y = f^{-1}(x)$. **3**

ii. Find an expression for $f^{-1}(x)$. **2**

iii. Evaluate $f^{-1}(2)$. **1**

(d) Use mathematical induction to show that for all positive integers $n \geq 2$, **3**

$$2 \times 1 + 3 \times 2 + 4 \times 3 + \cdots + n(n-1) = \frac{n(n^2-1)}{3}$$

Question 13 (15 Marks) Commence a NEW page. **Marks**

(a) i. Show that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$. **2**

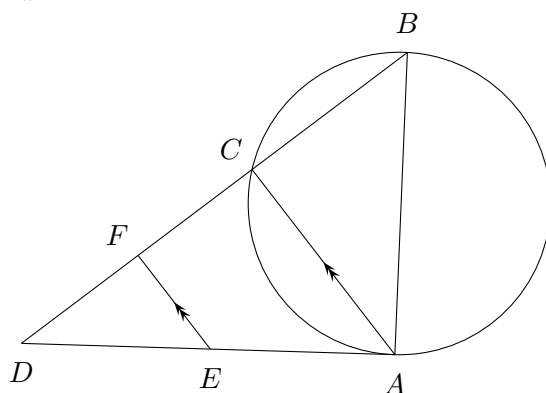
ii. Hence evaluate $\int 2 \sin^3 \theta \, d\theta$ **1**

(b) $P(x) = ax^3 - 7x^2 + kx + 4$ has $x - 4$ as a factor. When $P(x)$ is divided by $(x - 1)$, the remainder is -6 .

i. Determine the values of a and k . **2**

ii. Evaluate $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. **2**

(c) AB is a diameter of the circle and C is a point on the circle. The tangent to the circle at A meets BC produced at D . E is a point on AD and F is a point on CD such that $EF \parallel AC$.

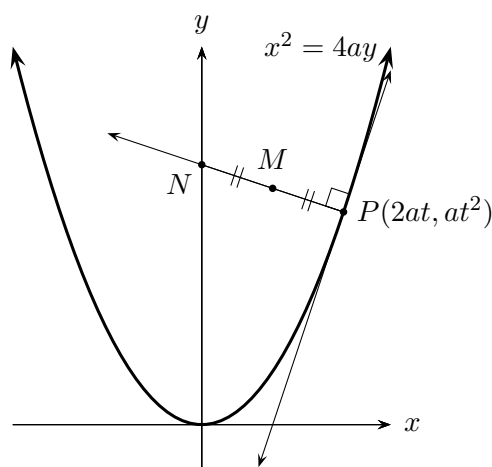


i. State why $\angle EAC = \angle ABC$. **1**

ii. Hence show that $EABF$ is a cyclic quadrilateral. **2**

iii. Show that BE is a diameter of the circle through E , A , B and F . **1**

(d) $P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$. The normal to the parabola at P cuts the y axis at N . M is the midpoint of PN .



i. Show that the normal has equation $x + ty = 2at + at^3$. **2**

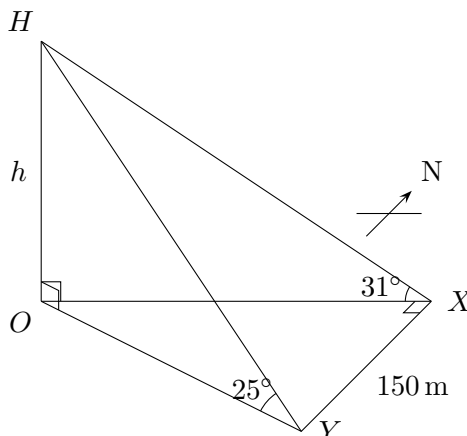
ii. Find the equation of the locus of M as P moves on the parabola. **2**

Question 14 (15 Marks)

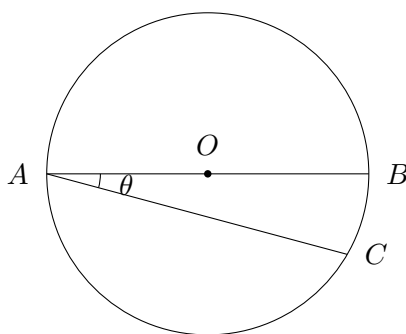
Commence a NEW page.

Marks

- (a) The radius of a spherical balloon is expanding at a constant rate of 6 cm s^{-1} .
- At what rate is the volume of the balloon expanding when its radius is 4 cm? **2**
 - At what rate is the surface area of the balloon expanding if the rate of change of volume is $15 \text{ cm}^3 \text{ s}^{-1}$? **3**
- (b) From a point X due east of a tower, the angle of elevation of the top of the tower H is 31° . From another point Y due south of X , the angle of elevation is 25° . $XY = 150 \text{ m}$.



- Show that $h = \frac{150 \tan 25^\circ \tan 31^\circ}{\sqrt{\tan^2 31^\circ - \tan^2 25^\circ}}$. **3**
 - Hence, find the height of the tower. **1**
- (c) AOB is the diameter of the circle with centre O and radius 1 metre. AC is a chord of the circle such that $\angle BAC = \theta$, where $0 < \theta < \frac{\pi}{2}$.



The area of the part of the circle contained between AB and AC is equal to one quarter of the area of the circle.

- Show that $\theta + \frac{1}{2} \sin 2\theta - \frac{\pi}{4} = 0$. **2**
- Show that $0.4 < \theta < 0.5$. **2**
- Use one application of Newton's Method with an initial approximation of $\theta_0 = 0.4$ to find the next approximation to θ , correct to 2 decimal places. **2**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

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- | | | | | |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☐ C | ☐ D |

Suggested Solutions

(2) $\div$ (1):

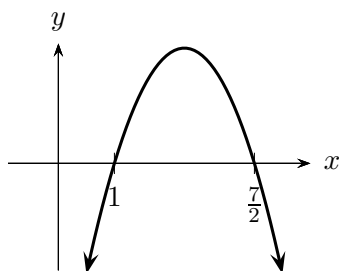
Section I

1. (D) 2. (B) 3. (D) 4. (A) 5. (B) 6. (A) 7. (C)
8. (C) 9. (D) 10. (D)

Question 11 (Ziaziaris)

(a) (2 marks)

$$\begin{aligned}\frac{5}{x-1} &> 2 \\ 5(x-1) &> 2(x-1)^2 \\ 5(x-1) - 2(x-1)^2 &> 0 \\ (x-1)(5-2(x-1)) &> 0 \\ (x-1)(7-2x) &> 0\end{aligned}$$



$$\therefore 1 < x < \frac{7}{2}$$

(b) (4 marks)

$$\begin{aligned}\sqrt{3} \cos \theta - \sin \theta &= 1 \\ &\equiv R \cos(\theta + \alpha) \\ &= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha \\ \begin{cases} R \cos \alpha = \sqrt{3} & (1) \\ R \sin \alpha = 1 & (2) \end{cases}\end{aligned}$$

$$(1)^2 + (2)^2:$$

$$\begin{aligned}R^2 &= 3 + 1 = 4 \\ \therefore R &= 2\end{aligned}$$

(c) i. (1 mark)

$$\begin{aligned}\tan \alpha &= \frac{1}{\sqrt{3}} \\ \therefore \alpha &= \frac{\pi}{6} \\ \therefore \sqrt{3} \cos \theta - \sin \theta &= 2 \cos \left(\theta + \frac{\pi}{3} \right) = 1 \\ \therefore \cos \left(\theta + \frac{\pi}{6} \right) &= \frac{1}{2} \\ \theta + \frac{\pi}{6} &= \frac{\pi}{3}, \frac{5\pi}{3} \\ \therefore \theta &= \frac{\pi}{6}, \frac{3\pi}{2}\end{aligned}$$

ii. (2 marks)

$$\begin{aligned}T &= 20 + Ae^{-kt} \\ \therefore T - 20 &= Ae^{-kt} \\ \frac{dT}{dt} &= -k \cdot Ae^{-kt} \\ &= -k(T - 20)\end{aligned}$$

$$T = 20 + Ae^{-kt}$$

When $t = 0$, $T = 150$:

$$\begin{aligned}150 &= 20 + Ae^0 \\ \therefore A &= 130 \\ \therefore T &= 20 + 130e^{-kt}\end{aligned}$$

When $t = 15$, $T = 100$

$$100 = 20 + 130e^{-15k}$$

$$\frac{80}{130} = e^{-15k}$$

$$-15k = \log_e \frac{8}{13}$$

$$k = -\frac{1}{15} \log_e \frac{8}{13} \approx 0.032 \text{ (3 dp)}$$

iii. (2 marks)

$$25 = 20 + 130e^{-0.032t}$$

$$\frac{5}{13} = e^{-0.032t}$$

$$-0.032t = \log_e \frac{5}{13}$$

$$t = \frac{1}{-0.032} \log_e \frac{5}{13}$$

$$\approx 100.66 \text{ min}$$

If exact values are used,

$$\begin{aligned} t &\approx 101.82 \text{ min} \\ &= 101 \text{ min } 49 \text{ s} \\ &\approx 102 \text{ min} \end{aligned}$$

(d) i. (1 mark)

$$\begin{aligned} v &= \sqrt{x} \\ \therefore v^2 &= x \\ \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} \left(\frac{1}{2} x \right) \\ &= \frac{1}{2} \end{aligned}$$

which is independent of x .

ii. (2 marks)

$$\frac{dx}{dt} = x^{\frac{1}{2}}$$

Separating variables,

$$\begin{aligned} \frac{dx}{x^{\frac{1}{2}}} &= dt \\ \int x^{-\frac{1}{2}} dx &= \int dt \\ \therefore t &= 2x^{\frac{1}{2}} + C \end{aligned}$$

When $t = 0$, $x = 1$.

$$\begin{aligned} 0 &= 2 + C \\ \therefore C &= -2 \\ \therefore t &= 2x^{\frac{1}{2}} - 2 \\ 2\sqrt{x} &= t + 2 \\ \therefore x &= \left(\frac{t+2}{2} \right)^2 \quad \left(= \frac{1}{4}t^2 + t + 1 \right) \end{aligned}$$

iii. (1 mark)

$$\begin{aligned} x(2) &= \left(\frac{2+2}{2} \right)^2 = 4 \\ x(3) &= \left(\frac{3+2}{2} \right)^2 = \frac{25}{4} \\ \text{Dist travelled} &= \frac{25}{4} - 4 = 2.25 \text{ m} \end{aligned}$$

Question 12 (Lin)

(a) (2 marks)

Let $t = \tan\left(\frac{\theta}{2}\right)$:

$$\begin{aligned} \frac{\sin x}{1 - \cos x} &= \frac{\frac{2t}{1+t^2}}{1 - \left(\frac{1-t^2}{1+t^2}\right)} \\ &= \frac{\frac{2t}{1+t^2}}{\frac{1+t^2-(1-t^2)}{1+t^2}} \\ &= \frac{2t}{2t^2} \\ &= \frac{1}{t} = \cot \frac{x}{2} \end{aligned}$$

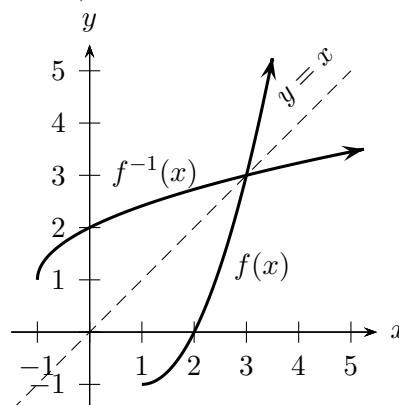
(b) (4 marks)

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin 2x}{1 + \sin^2 x} dx$$

Let $u = \sin^2 x$:

$$\begin{aligned} \frac{du}{dx} &= 2 \sin x \cos x = \sin 2x \\ \therefore du &= \sin 2x dx \\ x = \frac{\pi}{4} &\rightarrow u = \sin^2 \frac{\pi}{4} = \frac{1}{2} \\ x = \frac{\pi}{3} &\rightarrow u = \sin^2 \frac{\pi}{3} = \frac{3}{4} \\ \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin 2x}{1 + \sin^2 x} dx &= \int_{u=\frac{1}{2}}^{u=\frac{3}{4}} \frac{1}{1+u} du \\ &= \left[\log_e(1+u) \right]_{\frac{1}{2}}^{\frac{3}{4}} \\ &= \log_e \left(\frac{7}{4} \right) - \log_e \left(\frac{3}{2} \right) \\ &= \log_e \frac{\frac{7}{4}}{\frac{3}{2}} = \log_e \frac{7}{6} \end{aligned}$$

(c) i. (3 marks)



ii. (2 marks)

$$y = x^2 - 2x \quad x \geq 1$$

Interchanging variables,

$$x = y^2 - 2y \quad y \geq 1$$

$$x + 1 = y^2 - 2y + 1$$

$$\therefore (y - 1)^2 = x + 1$$

$$y - 1 = \sqrt{x + 1} \quad (\text{as } y \geq 1, \text{ take positive root})$$

$$\therefore y = 1 + \sqrt{x + 1}$$

iii. (1 mark)

$$f^{-1}(2) = 1 + \sqrt{2 + 1} = 1 + \sqrt{3}$$

(d) (3 marks)

Let $P(n)$ be the proposition

$$P(n) : \quad 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + n(n-1) \\ = \frac{n(n^2 - 1)}{3}$$

- Base case: $P(2)$ –

$$2(2 - 1) = 2$$

$$\frac{2(2^2 - 1)}{3} = \frac{2 \times 3}{3} = 2$$

Hence $P(2)$ is true.

- Inductive hypothesis: assume that $P(k)$ is true for some $k \in \mathbb{Z}^+$, i.e.

$$P(k) : \quad 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) \\ = \frac{k(k^2 - 1)}{3}$$

Now examine $P(k+1)$:

$$\begin{aligned} & \overbrace{2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1)}^{=P(k)} \\ & \quad + (k+1)([k+1] - 1) \\ & = \frac{k(k^2 - 1)}{3} + k(k+1) \\ & = \frac{k(k-1)(k+1) + 3k(k+1)}{3} \\ & = \frac{(k+1)(k^2 - k + 3k)}{3} \\ & = \frac{(k+1)(k^2 + 2k)}{3} \\ & = \frac{(k+1)(k^2 + 2k + 1 - 1)}{3} \\ & = \frac{(k+1)((k+1)^2 - 1)}{3} \end{aligned}$$

Question 13 (Berry)

(a) i. (2 marks)

$$\sin 3\theta = \sin(2\theta + \theta)$$

$$= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta$$

$$= 2 \sin \theta \cos \theta \cos \theta + (\cos^2 \theta - \sin^2 \theta) \sin \theta$$

$$= 2 \sin \theta \cos^2 \theta + \sin \theta \cos^2 \theta - \sin^3 \theta$$

$$= 2 \sin \theta (1 - \sin^2 \theta) + \sin \theta (1 - \sin^2 \theta) - \sin^3 \theta$$

$$= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - \sin^3 \theta - \sin^3 \theta$$

$$= 3 \sin \theta - 4 \sin^3 \theta$$

ii. (1 mark)

$$\begin{aligned} \int 2 \sin^3 \theta \, d\theta &= \frac{1}{2} \int 4 \sin^3 \theta \, d\theta \\ &= \frac{1}{2} \int (3 \sin \theta - \sin 3\theta) \, d\theta \\ &= \frac{1}{2} \left[-3 \cos \theta + \frac{1}{3} \cos 3\theta \right] + C \\ &= \frac{1}{6} \cos 3\theta - \frac{3}{2} \cos \theta + C \end{aligned}$$

(b) i. (2 marks)

$$P(x) = ax^3 - 7x^2 + k(4) + 4$$

As $(x - 4)$ is a factor, by the factor theorem:

$$P(4) = 0$$

$$\therefore a \times (4^3) - 7(4^2) + 4k + 4 = 0$$

$$64a - 112 + 4k + 4 = 0$$

$$16a + k - 27 = 0$$

Using the remainder theorem,

$$P(1) = -6$$

$$\therefore a(1^3) - 7(1^2) + k(1) + 4 = -6$$

$$a - 7 + k + 4 = -6$$

$$a + k - 3 = -6$$

$$k = -3 - a$$

$$\begin{cases} 16a + k = 27 & (1) \\ k = -3 - a & (2) \end{cases}$$

Substitute (2) to (1):

$$\begin{aligned} 16a + (-3 - a) &= 27 \\ 15a &= 30 \\ a &= 2 \\ \therefore k &= -5 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \\ &= \frac{\left(\frac{c}{a}\right)}{\left(-\frac{d}{a}\right)} = \frac{\left(\frac{-5}{2}\right)}{\left(-\frac{4}{2}\right)} \\ &= \frac{5}{4} \end{aligned}$$

(c) i. (1 mark)

The angle between a tangent and a chord through the point of contact is equal to the angle in the alternate segment.

ii. (2 marks)

- $\angle AEF = 180^\circ - \angle EAC$
(Cointerior angles, $AC \parallel EF$)
- $\angle AEF = 180^\circ - \angle ABC$
(Since $\angle EAC = \angle ABC$)

iii. (1 mark)

- $\angle EAB = 90^\circ$ (The tangent to a circle is perpendicular to the diameter drawn to the point of contact)
- $\therefore BE$ is the diameter of the circle through E , A , B and F as it subtends a right angle at the circumference of the circle at point A .

(d) i. (2 marks)

$$\begin{aligned} x^2 &= 4ay \\ y &= \frac{x^2}{4a} \\ \frac{dy}{dx} &= \frac{2x}{4a} = \frac{x}{2a} \end{aligned}$$

At the point $P(2at, at^2)$

$$\begin{aligned} m_{\text{tan}} &= \frac{2at}{2a} = t \\ \therefore m_{\perp} &= -\frac{1}{t} \end{aligned}$$

Apply the point-gradient formula,

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - at^2 &= -\frac{1}{t}(x - 2at) \\ ty - at^3 &= -x + 2at \\ x + ty &= 2at + at^3 \end{aligned}$$

ii. (2 marks)

Find point N by letting $x = 0$:

$$\begin{aligned} 0 + ty &= 2at + at^3 \\ \therefore y &= 2a + at^2 \quad (t \neq 0) \end{aligned}$$

Finding the midpoint M of PN

$$\begin{aligned} M &= \left(\frac{0 + 2at}{2}, \frac{2a + at^2 + at^2}{2} \right) \\ &= \left(\frac{2at}{2}, \frac{2a + 2at^2}{2} \right) \\ &= (at, a + at^2) \\ \begin{cases} x = at & (1) \\ y = a + at^2 & (2) \end{cases} \end{aligned}$$

Substitute (1) into (2) after rearranging,

$$\begin{aligned} y &= a + a \left(\frac{x}{a} \right)^2 \\ &= a + \frac{x^2}{a} \\ \therefore ay &= a^2 + x^2 \\ x^2 &= ay - a^2 = a(y - a) \end{aligned}$$

Question 14 (Lam)

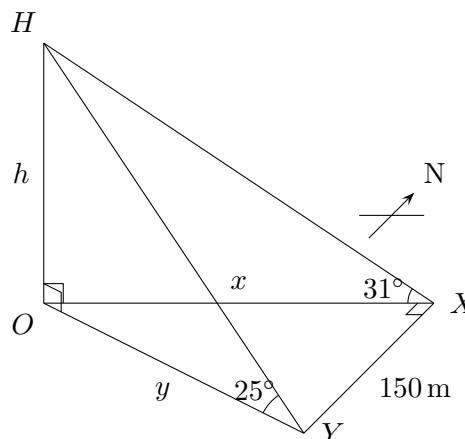
(a) i. (2 marks)

$$\left| \begin{aligned} \frac{dV}{dt} &= \frac{dV}{dr} \times \frac{dr}{dt} \\ V &= \frac{4}{3}\pi r^3 \quad \frac{dr}{dt} = 6 \text{ cms}^{-1} \\ \frac{dV}{dt} &= 4\pi r^2 \\ \frac{dV}{dt} &= 4\pi r^2 \times 6 \Big|_{r=4} \\ &= 4 \times \pi \times 4^2 \times 6 \\ &= 384\pi \end{aligned} \right.$$

ii. (3 marks)

$$\begin{aligned} A &= 4\pi r^2 \quad \frac{dA}{dr} = 8\pi r \\ \frac{dV}{dt} &= \frac{dV}{dr} \times \frac{dr}{dt} \\ 15 &= 4\pi r^2 \times 6 \\ \frac{15}{24\pi} &= r^2 \\ \therefore r &= \sqrt{\frac{5}{8\pi}} \\ \frac{dA}{dt} &= \frac{dA}{dr} \times \frac{dr}{dt} \\ &= 8\pi r \times 6 \\ &= 8\pi \times \sqrt{\frac{5}{8\pi}} \times 6 \\ &= \sqrt{1440\pi} \\ &= 67.3 \text{ cm}^2\text{s}^{-1} \text{ (1 dp)} \end{aligned}$$

(b) i. (3 marks)

• In $\triangle HOX$

$$\begin{aligned} \frac{h}{x} &= \tan 31^\circ \\ \therefore x &= \frac{h}{\tan 31^\circ} \end{aligned}$$

• Similarly in $\triangle HOY$,

$$\begin{aligned} \frac{h}{y} &= \tan 25^\circ \\ \therefore y &= \frac{h}{\tan 25^\circ} \end{aligned}$$

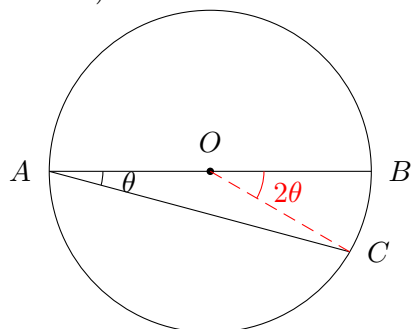
• In $\triangle OXY$,

$$\begin{aligned} x^2 + 150^2 &= y^2 \\ \left(\frac{h}{\tan 31^\circ}\right)^2 + 150^2 &= \left(\frac{h}{\tan 25^\circ}\right)^2 \\ \left(\frac{h}{\tan 25^\circ}\right)^2 - \left(\frac{h}{\tan 31^\circ}\right)^2 &= 150^2 \\ h^2 \left(\frac{1}{\tan^2 25^\circ} - \frac{1}{\tan^2 31^\circ}\right) &= 150^2 \\ h^2 \left(\frac{\tan^2 31^\circ - \tan^2 25^\circ}{\tan^2 25^\circ \tan^2 31^\circ}\right) &= 150^2 \\ h^2 &= \frac{150^2 \tan^2 25^\circ \tan^2 31^\circ}{\tan^2 31^\circ - \tan^2 25^\circ} \\ \therefore h &= \frac{150 \tan 25^\circ \tan 31^\circ}{\sqrt{\tan^2 31^\circ - \tan^2 25^\circ}} \end{aligned}$$

ii. (1 mark)

$$h = 110.91 \text{ m (2 dp)}$$

(c) i. (2 marks)



$\angle COB = 2\theta$ as angle at the centre is twice the angle at the circumference, subtended by the same arc. (Can also use angle sum of $\triangle$ and exterior $\angle$)

$$A_{\text{sector } OBC} + A_{\triangle OAC} = \frac{1}{4}\pi r^2$$

$$\begin{aligned} A_{\text{sector } OBC} &= \frac{1}{2}r^2 \times (2\theta) \\ &= r^2\theta = \theta \quad (r = 1) \end{aligned}$$

$$\begin{aligned} A_{\triangle OBC} &= \frac{1}{2}r^2 \times \sin(\pi - 2\theta) \\ &= \frac{1}{2}r^2 \sin 2\theta = \frac{1}{2} \sin 2\theta \end{aligned}$$

(As $\sin(\pi - x) = \sin x$)

Adding areas and equating,

$$\begin{aligned} \theta + \frac{1}{2} \sin 2\theta &= \frac{\pi}{4} \\ \therefore \theta + \frac{1}{2} \sin 2\theta - \frac{\pi}{4} &= 0 \end{aligned}$$

ii. (2 marks)

- When $\theta = 0.4$,

$$0.4 + \frac{1}{2} \sin 0.8 - \frac{\pi}{4} \approx -0.026 \dots$$

- When $\theta = 0.5$,

$$0.5 + \frac{1}{2} \sin 1 - \frac{\pi}{4} \approx 0.135$$

If $f(\theta) = \theta + \frac{1}{2} \sin 2\theta - \frac{\pi}{4}$, then $f(\theta)$ is continuous for all $x \in \mathbb{R}$ as θ and $\sin 2\theta$ are both continuous. From above, $f(0.4) < 0$ and $f(0.5) > 0$, therefore a zero crossing exists for $0.4 < \theta < 0.5$, satisfying the equation found in part (i).

iii. (2 marks)

$$\begin{aligned} &\left| \begin{aligned} f(\theta) &= \theta + \frac{1}{2} \sin 2\theta - \frac{\pi}{4} \\ f'(\theta) &= 1 + \frac{1}{2} \times 2 \cos 2\theta \\ &= 1 + \cos 2\theta \end{aligned} \right. \\ \theta_1 &= \theta_0 - \frac{f(\theta_0)}{f'(\theta_0)} \\ &= 0.4 - \frac{0.4 + \frac{1}{2} \sin 0.8 - \frac{\pi}{4}}{1 + \cos 0.8} \\ &\left(= 0.4 - \frac{-0.02672011 \dots}{1.696706 \dots} \right) \\ &= 0.41574 \dots \\ &= 0.42 \text{ (2 dp)} \end{aligned}$$